The LNM Institute of Information Technology, Jaipur
Software Development Laboratory
Course Code: CS314
Lab Assignment 1: Solution

1. Project Identification

Item	Details
Project Name	Transcript Generation System (TGS)
Project Type	Web-based Academic Transcript Generation & Verification System
Team ID	T029
Team Members	Ayush Agarwal, Yash Sharma, Jatin Dangi, Daksh Kedia
Roll Numbers	24UCC171, 24UCC172, 24UCC173, 24UCC174
Development Model	Agile + Incremental Development
Primary Users	Students, Event Coordinators, Administrators
Development Team	Student software development team
Course	Software Engineering
Project Objective	To develop a system that generates verified academic transcripts by calculating course credits, grades, semester (SGPA) and cumulative (CGPA) performance from institutionally verified academic records
Expected Outcome	A working, tested and documented software system developed incrementally

2. Project Background

Current Problem

The current process for generating academic transcripts relies heavily on manual data entry, creating significant bottlenecks for both university administrators and students. Examination offices typically juggle separate spreadsheets, paper files, and scanned mark sheets for every individual semester.

This fragmented approach causes two primary issues:

- **Administrative Inefficiency:** Manually calculating earned credits, SGPAs, and CGPAs across hundreds of students is tedious. It frequently introduces human error, miscalculations, and formatting inconsistencies.
- **Student Delays:** When students need transcripts for higher education, internships, or job placements, they face frustrating wait times. Staff must manually track down, compile, and double-check multiple semester records before issuing the final document.

Automating this workflow is essential to eliminate calculation errors, reduce staff workload, and ensure students receive their critical academic credentials without unnecessary administrative delays

Proposed Solution

The TGS provides a centralized, reliable platform designed to automate the calculation of earned credits, SGPAs, and CGPAs from institutionally verified records, ensuring that consistently formatted and approved academic transcripts are easily generated and managed for students.

3. Project Vision

The TGS provides a centralized, reliable platform designed to automate the calculation of earned credits, SGPAs, and CGPAs from institutionally verified records, ensuring that consistently formatted and approved academic transcripts are easily generated and managed for students.

4. Problem Statement

Institutes need to issue academic transcripts to students for purposes such as higher education, placements and scholarships. Existing practices, which often rely on spreadsheets and manual verification across departments, are susceptible to calculation errors, formatting inconsistencies and processing delays. TGS is proposed as an integrated software solution that stores verified academic records, automates credit, SGPA and CGPA calculation according to a fixed grading scheme, and generates the final transcript only after appropriate verification, thereby reducing manual effort and error while improving turnaround time.

5. Project Objectives

The initial objectives should be:

- To provide a centralized platform for storing and managing verified semester-wise academic records of students.
- To automatically calculate credits earned for each course based on the grade obtained.
- To automatically calculate the Semester Grade Point Average (SGPA) for each completed semester using the defined grading formula.
- To automatically calculate the Cumulative Grade Point Average (CGPA) across all completed semesters.
- To provide role-based access for students, course coordinators, academic office staff and administrators.
- To implement a verification and approval workflow so that a transcript is marked official only after review.
- To generate a properly formatted, downloadable transcript (PDF) for a student on request.
- To maintain an audit trail of record entries, edits and approvals for accountability.
- To allow the academic office to securely upload or update semester grade records.
- To develop the system incrementally using standard software engineering practices.

6. Stakeholders

Stakeholder	Interest / Role
Stakeholder	Interest / Role
Students	View, download and verify their own academic transcript
Academic / Examination Office	Upload and verify semester-wise grade records; approve transcripts
Course Coordinators / Faculty	Submit and confirm grade data for their courses
Administrator	Manage users, roles, grading scheme and overall system configuration
Institute Management	Monitor academic performance and obtain consolidated reports
System Development Team	Design, develop, test and maintain the Transcript Generation System

7. Initial Scope

TGS will initially cover:

- User registration/login
- Role-based access (student, faculty, academic office, administrator)
- Semester-wise course and grade record entry/upload
- Credit calculation engine based on grade and course credit weight
- SGPA calculation per semester
- CGPA calculation across completed semesters
- Record verification and approval workflow
- Transcript generation and download (PDF)
- Transcript/record history view for students
- Search and filter of student academic records (for staff)
- Basic reports/dashboard for the academic office

Initially Out of Scope

To keep the project manageable, we should initially exclude things such as:

- Online fee/payment processing
- Online examination or exam-conduct features
- Course registration/enrollment management
- Complete ERP functionality
- Backlog/re-evaluation and grade-revision workflows beyond basic correction
- Mobile native application
- Integration with external universities or degree-verification agencies
- AI-based performance prediction or analytics

8. Initial Assumptions

- Users will have access to the Internet.
- Users will have valid institutional accounts or credentials.
- Grade records supplied by the academic office are accurate and verified before entry into the system.
- The grading scale and credit structure (e.g., grade points per letter grade) are defined and fixed in advance.
- The system will initially be developed as a web application.
- The system will use a centralized database.
- Only one academic program/curriculum structure is handled during the initial version.

9. Initial Constraints

- Limited development time
- Student development team
- Limited access to real institutional academic data; sample/mock data will be used for development and testing.
- Limited infrastructure available during development.
- Possible requirement changes during development.
- Need to protect sensitive student academic data and ensure privacy.
- Limited testing opportunities with real end users.
- Grading and credit policies may vary across departments; the initial version assumes a single, uniform scheme.

10. Initial Success Criteria

TGS will be considered successful if the team can demonstrate that:

1. different user roles (student, faculty, academic office, administrator) can access only their appropriate functionality;
2. the academic office can upload or enter semester-wise grade records for a student;
3. the system correctly calculates credits earned and SGPA for a given semester;
4. the system correctly calculates CGPA across multiple completed semesters;
5. a grade record can pass through the verification/approval workflow before being marked official;
6. a verified transcript can be generated and downloaded as a properly formatted PDF;
7. unauthorized users cannot access another student's academic records;
8. important system functions (credit/SGPA/CGPA calculation, access control) are tested;
9. defects are documented and resolved;
10. the project is maintained in a controlled Git/GitHub repository with meaningful commit history.

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Lab Assignment 2: Solution

11. User Classes

11.1 Student

Students are enrolled individuals who need academic transcripts for various purposes such as higher education, placements, and scholarships.

They shall be able to:

- Authenticate using institutional credentials
- View their own academic records
- View semester-wise grades and credits
- View calculated SGPA and CGPA
- Request transcript generation
- Download generated transcripts
- View transcript generation history
- Receive notifications about transcript status

11.2 Course Coordinator / Faculty

Course coordinators and faculty members are responsible for submitting and confirming grade data for courses they teach.

They shall be able to:

- Log in with authorized credentials
- Submit course grades for students
- Confirm submitted grade data
- Update grades within permitted timelines
- View submitted grade records
- Receive confirmation of grade submission

11.3 Academic / Examination Office Staff

Academic/Examination office staff manage the verification and approval of academic records and transcripts.

They shall be able to:

- Log in with authorized credentials

- Upload semester-wise grade records
- Verify student academic records
- Approve transcript generation requests
- Search and filter student records
- Correct errors in academic records subject to proper authorization
- View pending verification/approval requests
- Generate academic reports
- Access audit trail of record modifications

11.4 Administrator

The system administrator manages users, system configuration, and overall system operations.

The administrator shall:

- Manage user accounts across all roles
- Assign and modify user roles
- Configure grading scheme and credit structure
- Manage system-level settings
- Monitor system activity and logs
- Access comprehensive administrative reports
- Maintain audit trail records
- Handle system backup and security

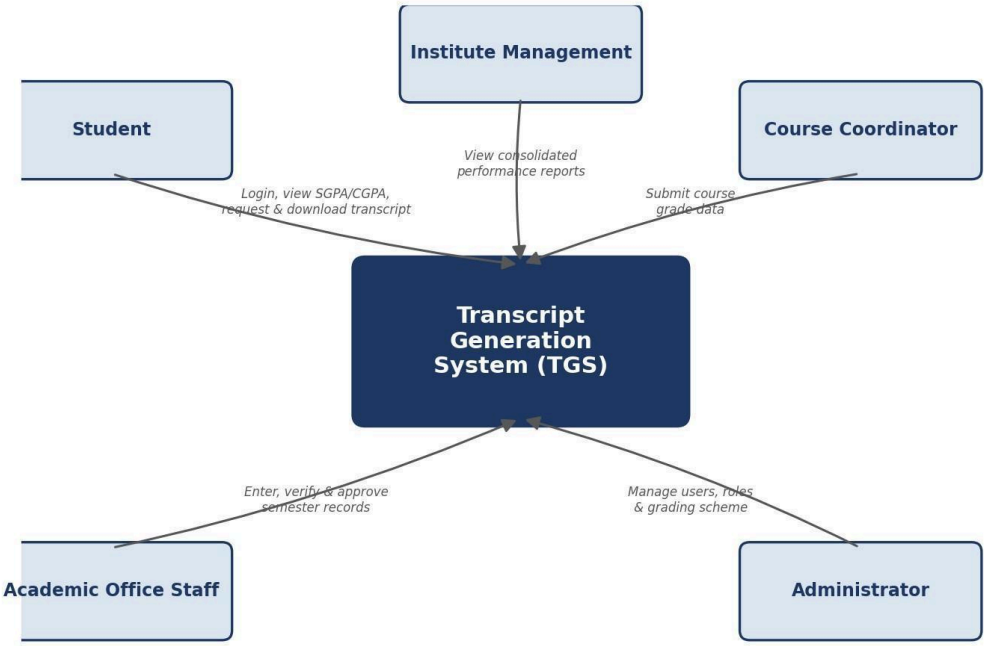
11.5 Institute Management

Management users require aggregated academic performance data for institutional oversight.

They shall have access to:

- Consolidated academic performance reports
- Batch-wise and department-wise analytics
- SGPA and CGPA distribution reports
- Transcript generation statistics

12. System Context



13. Functional Requirements

13.1 Authentication and Account Management

FR-001 — User Login
The system shall allow registered users to authenticate using valid institutional credentials.

FR-002 — Authentication Failure Handling
The system shall notify users when authentication fails without revealing sensitive security information.

FR-003 — Logout
The system shall allow authenticated users to terminate their session securely.

FR-004 — Role Assignment
The system shall allow authorized administrators to assign roles to users.

FR-005 — Role-Based Access Control
The system shall restrict system functions according to the user's assigned role.

13.2 Grade Record Management

FR-006 — Upload Grade Records
The system shall allow authorized academic office staff to upload semester-wise grade records for students.

FR-007 — Submit Course Grades
The system shall allow authorized course coordinators to submit grades for students enrolled in their courses.

FR-008 — Grade Data Entry
The system shall allow entry of grade information including:

- Student ID
- Course code
- Course name
- Credits
- Grade obtained
- Semester
- Academic year

FR-009 — Edit Grade Records
The system shall allow authorized users to edit grade records subject to defined permissions and approval workflows.

FR-010 — Grade Record Verification
The system shall allow authorized academic office staff to verify submitted grade records before finalization.

FR-011 — Grade Record Approval
The system shall require approval from authorized personnel before grade records become official.

13.3 Credit Calculation

FR-012 — Calculate Credits Earned
The system shall automatically calculate credits earned for each course based on the grade obtained and the course credit weight.

FR-013 — Grade Point Mapping
The system shall use a predefined grading scheme to map letter grades to grade points.

FR-014 — Credit Validation
The system shall validate that only passing grades earn credits according to the defined grading rules.

FR-015 — Display Earned Credits
The system shall display total credits earned per semester and cumulatively.

13.4 SGPA Calculation

FR-016 — Calculate SGPA
The system shall automatically calculate the Semester Grade Point Average (SGPA) for each

completed semester using the formula:
SGPA = Σ(Grade Points × Credits) / Σ(Credits)

FR-017 — Display SGPA
The system shall display the calculated SGPA for each semester on the student's academic record.

FR-018 — SGPA Recalculation
The system shall recalculate SGPA automatically when grade records are updated or corrected.

13.5 CGPA Calculation

FR-019 — Calculate CGPA
The system shall automatically calculate the Cumulative Grade Point Average (CGPA) across all completed semesters using the formula:
CGPA = Σ(Grade Points × Credits for all semesters) / Σ(Credits for all semesters)

FR-020 — Display CGPA
The system shall display the current CGPA on the student's academic record.

FR-021 — CGPA Recalculation
The system shall recalculate CGPA automatically when any semester grade records are updated.

13.6 Transcript Generation

FR-022 — Request Transcript
The system shall allow authenticated students to request generation of their academic transcript.

FR-023 — Transcript Content
The system shall include in the generated transcript:

- Student personal information
- Semester-wise course details (code, name, credits, grade)
- Semester-wise SGPA
- Cumulative CGPA
- Total credits earned
- Transcript generation date
- Verification status

FR-024 — Transcript Format
The system shall generate transcripts in a standardized, professionally formatted PDF document.

FR-025 — Transcript Approval Workflow
The system shall require academic office approval before a transcript is marked as official.

FR-026 — Download Transcript
The system shall allow students to download their approved transcript as a PDF file.

FR-027 — Transcript Watermark
The system shall apply appropriate watermarks indicating the transcript's official status.

13.7 Record Viewing and Access

FR-028 — View Own Academic Records
The system shall allow students to view their complete academic records including all semesters.

FR-029 — View Transcript History
The system shall allow students to view their transcript generation history including request dates and status.

FR-030 — Search Student Records
The system shall allow authorized academic office staff to search student records by student ID, name, or batch.

FR-031 — Filter Student Records
The system shall allow authorized staff to filter student records by relevant criteria such as program, semester, or year.

13.8 Notification Management

FR-032 — Grade Submission Notification
The system shall notify students when new grades are submitted or updated.

FR-033 — Transcript Status Notification
The system shall notify students about the status of their transcript request (pending, approved, ready for download).

FR-034 — Approval Notification
The system shall notify academic office staff when transcript approval is required.

FR-035 — View Notifications
The system shall allow users to view their notifications.

13.9 Reporting and Dashboard

FR-036 — Academic Office Dashboard
The system shall provide academic office staff with a dashboard showing pending approvals, recent transcript requests, and key statistics.

FR-037 — Student Performance Report
The system shall generate reports showing student academic performance including SGPA and CGPA trends.

FR-038 — Batch Performance Report
The system shall generate batch-wise performance reports for authorized users.

FR-039 — Transcript Generation Report
The system shall generate reports on transcript generation activity and statistics.

FR-040 — Export Reports
The system shall allow authorized users to export reports in standard formats (PDF, CSV).

13.10 Audit Trail

FR-041 — Record Modifications Log
The system shall maintain an audit trail of all grade record entries, edits, and approvals.

FR-042 — User Action Log
The system shall log significant user actions including who made changes, what was changed, and when.

FR-043 — View Audit Trail
The system shall allow authorized administrators to view the audit trail for accountability and troubleshooting.

14. Business Rules

ID	Business Rule
BR-001	Only authorized academic office staff can upload or modify grade records.
BR-002	Grade records must be verified before being used in transcript generation.
BR-003	Credits are earned only for grades above the minimum passing grade as defined in the grading scheme.
BR-004	SGPA is calculated only for semesters with complete and verified grade records.
BR-005	CGPA includes all completed semesters with verified grades.
BR-006	A transcript can be marked official only after approval by authorized academic office staff.
BR-007	Students can only view and download their own academic records and transcripts.
BR-008	The grading scheme and credit structure must be configured before grade entry.
BR-009	Grade modifications after approval require additional authorization and must be audited.

BR-010	Only students with at least one completed semester can request a transcript.
BR-011	Course coordinators can only submit grades for courses assigned to them.
BR-012	All transcript generation requests must be logged with timestamp and requester information.

15. Non-Functional Requirements

15.1 Performance

NFR-001: The system should provide acceptable response times for grade record retrieval and SGPA/CGPA calculation under normal load.

NFR-002: Transcript generation should complete within 10 seconds for a typical 8-semester academic record.

NFR-003: The system should support concurrent access by at least 100 users during peak periods such as semester end.

15.2 Security

NFR-004: All system functionality shall require user authentication.

NFR-005: Access to academic records and transcripts shall be controlled strictly by user roles.

NFR-006: Student academic data shall not be accessible to unauthorized users.

NFR-007: All grade record modifications shall be logged with user identity, timestamp, and modification details.

NFR-008: Passwords shall be stored using industry-standard hashing algorithms.

NFR-009: Transcript PDFs shall include security features to prevent unauthorized modification.

15.3 Usability

NFR-010: The interface shall provide clear navigation for accessing academic records and requesting transcripts.

NFR-011: Error messages shall be informative and guide users toward resolution.

NFR-012: Common tasks such as viewing grades and downloading transcripts should require minimal steps.

NFR-013: The system should provide helpful guidance for first-time users.

15.4 Reliability

NFR-014: Grade records and calculated SGPA/CGPA values shall be maintained reliably without loss.

NFR-015: Calculation of credits, SGPA, and CGPA must be accurate according to the defined formulae.

NFR-016: The system shall handle errors gracefully and inform users appropriately.

15.5 Availability

NFR-017: The system should be available 99% of the time during academic office hours, excluding scheduled maintenance.

15.6 Maintainability

NFR-018: The software shall follow a modular architecture separating grade management, calculation engine, and transcript generation.

NFR-019: Source code shall be maintained in a version-controlled Git repository.

NFR-020: Documentation shall be maintained alongside system changes.

15.7 Scalability

NFR-021: The system architecture should support growth to accommodate thousands of student records across multiple academic years.

NFR-022: The database design should allow efficient querying as data volume increases.

15.8 Compatibility

NFR-023: The system should operate correctly on commonly used modern web browsers (Chrome, Firefox, Safari, Edge).

NFR-024: Generated transcript PDFs should be viewable on all standard PDF readers.

15.9 Privacy

NFR-025: Student academic information shall be accessible only to the student themselves and authorized staff.

NFR-026: The system shall comply with institutional data protection policies.

15.10 Auditability

NFR-027: All grade record entries, modifications, and approvals should be recorded in an audit trail.

NFR-028: Transcript generation requests and approvals should be logged for future reference.

16. Data Requirements

The initial logical information requirements include the following entities:

User

Possible attributes:

- User ID
- Name
- Email
- Role (Student, Faculty, Academic Office, Administrator)
- Department/Program
- Account status
- Last login

Student

Possible attributes:

- Student ID
- Name
- Email
- Program/Branch
- Batch/Year of admission
- Current semester
- Account status

Course

Possible attributes:

- Course ID/Code
- Course name
- Credits
- Department
- Course type (Core, Elective, etc.)

Grade Record

Possible attributes:

- Record ID
- Student ID
- Course code
- Semester
- Academic year
- Grade obtained
- Credits assigned
- Credits earned
- Submission date
- Verification status
- Approved by
- Approval date

Grading Scheme

Possible attributes:

- Grade (A+, A, B+, B, C, D, F, etc.)
- Grade points
- Minimum percentage/marks
- Pass/Fail status

Semester Record

Possible attributes:

- Record ID
- Student ID
- Semester number
- Academic year
- Total credits registered
- Total credits earned
- SGPA
- Calculation date
- Status

Transcript

Possible attributes:

- Transcript ID
- Student ID
- Generation date
- Requested by
- Status (Pending, Approved, Generated)

- Approved by
- Approval date
- File path/URL
- Transcript type (Official, Unofficial)

Notification

Possible attributes:

- Notification ID
- Recipient (User ID)
- Notification type
- Message content
- Related entity (Grade Record, Transcript)
- Timestamp
- Read status

Audit Log

Possible attributes:

- Log ID
- User ID
- Action performed
- Entity affected (Grade Record, Transcript)
- Entity ID
- Changes made
- Timestamp
- IP address

17. Semester Record Lifecycle

A semester record may pass through the following states:

- Draft — created by academic office staff or a course coordinator, not yet submitted.
- Submitted — sent forward for verification.
- Under Verification — reviewed by academic office staff for correctness and completeness.
- Verified — confirmed correct, awaiting final approval.
- Approved — marked official; included in SGPA/CGPA and transcript calculation.
- Rejected — returned with a reason; may be revised and resubmitted.

18. Major Use Cases

The initial use-case catalogue is as follows:

ID	Use Case	Primary Actor
UC-001	Login	All users
UC-002	Manage Users	Administrator
UC-003	Configure Grading Scheme	Administrator
UC-004	Upload Grade Records	Academic Office Staff
UC-005	Submit Course Grades	Course Coordinator
UC-006	Verify Grade Records	Academic Office Staff
UC-007	View Academic Records	Student
UC-008	Calculate SGPA	System
UC-009	Calculate CGPA	System
UC-010	Request Transcript	Student
UC-011	Approve Transcript	Academic Office Staff
UC-012	Generate Transcript PDF	System
UC-013	Download Transcript	Student
UC-014	Search Student Records	Academic Office Staff
UC-015	View Audit Trail	Administrator
UC-016	Generate Reports	Academic Office Staff / Management
UC-017	View Notifications	All authenticated users

19. Detailed Use Case Example

UC-007 — View Academic Records

Primary Actor: Student

Preconditions:

- Student is authenticated
- Student has at least one semester with grade records

Main Flow:

1. Student navigates to "My Academic Records" section
2. System retrieves student's complete academic history
3. System displays semester-wise grade records
4. System displays courses with course code, name, credits, and grade
5. System displays calculated SGPA for each semester
6. System displays overall CGPA
7. System displays total credits earned
8. Student can view detailed information for any semester

Alternative Flows:

- If no grade records exist → System displays message that no records are available
- If SGPA/CGPA calculation is pending → System shows status as "Under Processing"

Postcondition:
Student has viewed their complete academic records.

UC-010 — Request Transcript

Primary Actor: Student

Preconditions:

- Student is authenticated
- Student has at least one completed semester with verified grades

Main Flow:

1. Student navigates to "Request Transcript" section
2. System displays transcript request form
3. Student selects transcript type (Official/Unofficial)
4. Student optionally provides purpose or remarks
5. Student submits transcript request

6. System validates that grade records are complete and verified
7. System creates transcript request record
8. System calculates final SGPA and CGPA
9. System generates preliminary transcript
10. System sends request to academic office for approval
11. System displays confirmation to student with request ID
12. System sends notification to academic office staff

Alternative Flows:

- If grade records are incomplete → System notifies student that transcript cannot be generated
- If student has pending corrections → System notifies student to resolve issues first
- If system error occurs during generation → System logs error and notifies administrators

Postcondition:
Transcript request is created and pending approval.

UC-011 — Approve Transcript

Primary Actor: Academic Office Staff

Preconditions:

- Staff member is authenticated with Academic Office role
- Transcript request exists and is pending approval
- All grade records are verified

Main Flow:

1. Staff member accesses pending transcript approval queue
2. System displays list of pending transcript requests
3. Staff member selects a transcript request to review
4. System displays student details and generated transcript preview
5. Staff member reviews transcript for accuracy and completeness
6. Staff member verifies student eligibility
7. Staff member approves the transcript request
8. System updates transcript status to "Approved"
9. System applies official watermark to transcript PDF
10. System records approval details (approver ID, timestamp)
11. System sends notification to student that transcript is ready
12. System logs the approval action in audit trail

Alternative Flows:

- If discrepancies are found → Staff member rejects request with comments
- If additional verification needed → Staff member marks request for manual review
- If student is not eligible → Staff member rejects request with reason

Postcondition:
Transcript is officially approved and ready for download by the student.

20. User Stories

US-001
As a student, I want to log in using my institutional credentials so that I can securely access my academic records.

US-002
As a student, I want to view my semester-wise grades and credits so that I can track my academic progress.

US-003
As a student, I want to see my calculated SGPA and CGPA so that I know my overall academic performance.

US-004
As a student, I want to request an academic transcript so that I can apply for higher education or job opportunities.

US-005
As a student, I want to download my approved transcript as a PDF so that I can submit it to external organizations.

US-006
As a course coordinator, I want to submit grades for my courses so that student records are updated.

US-007
As academic office staff, I want to upload semester-wise grade records so that transcripts can be generated.

US-008
As academic office staff, I want to verify grade records before finalization so that only accurate data is used.

US-009
As academic office staff, I want to approve transcript requests so that only validated transcripts are issued.

US-010
As academic office staff, I want to search and filter student records so that I can quickly find specific information.

US-011
As an administrator, I want to configure the grading scheme so that credit and GPA calculations are accurate.

US-012
As an administrator, I want to manage user roles so that access to the system is properly controlled.

US-013
As an administrator, I want to view the audit trail so that I can track system changes and ensure accountability.

US-014
As institute management, I want to generate academic performance reports so that I can monitor institutional outcomes.

US-015
As a student, I want to receive notifications when my transcript is ready so that I can download it promptly.

21. Acceptance Criteria Example

US-003 — View Calculated SGPA and CGPA

ID	Acceptance Criterion
AC-01	Student must be authenticated
AC-02	At least one semester must have verified grades
AC-03	SGPA must be displayed for each completed semester
AC-04	SGPA must be calculated using the formula: Σ(Grade Points × Credits) / Σ(Credits)
AC-05	CGPA must be displayed showing cumulative performance
AC-06	CGPA must include all completed semesters
AC-07	Calculations must be accurate to two decimal places
AC-08	System must show total credits earned

US-004 — Request Academic Transcript

ID	Acceptance Criterion
AC-01	Student must be authenticated
AC-02	Student must have at least one completed semester
AC-03	All grade records must be verified
AC-04	System must validate completeness of records before accepting request
AC-05	Request must be logged with timestamp and student ID
AC-06	Student must receive confirmation with request ID
AC-07	Academic office must be notified of pending approval
AC-08	Transcript must include all required information (grades, SGPA, CGPA, credits)

US-009 — Approve Transcript Requests

ID	Acceptance Criterion
AC-01	User must be authenticated as academic office staff
AC-02	Pending transcript requests must be displayed in a queue
AC-03	Staff must be able to preview the generated transcript
AC-04	Staff must be able to approve or reject the request
AC-05	Rejection must require a reason or comment
AC-06	Approved transcripts must be marked as "Official"
AC-07	Student must be notified upon approval
AC-08	Approval action must be logged in audit trail with staff ID and timestamp

22. Requirements Prioritization

Must Have

- User authentication and role-based access
- Grade record upload and entry
- Credit calculation based on grades
- SGPA calculation per semester
- CGPA calculation across semesters
- Transcript generation (PDF)
- Transcript approval workflow
- View academic records for students
- Basic notifications

Should Have

- Grade record verification workflow
- Search and filter student records
- Transcript history and status tracking
- Academic office dashboard
- Audit trail for record modifications
- Grade submission by course coordinators
- Basic academic performance reports

Could Have

- Advanced analytics and performance trends
- Batch-wise comparative reports
- Email notifications for all events
- Transcript request tracking with detailed status
- Automated reminders for pending approvals
- Mobile-responsive design
- Export functionality for reports

Won't Have in Initial Release

- Online fee payment for transcript requests
- Integration with external degree verification agencies
- Backlog/re-evaluation management workflows
- Complete ERP integration
- AI-based performance prediction
- Mobile native application
- Blockchain-based transcript verification
- Multiple curriculum structure support

23. Requirements Traceability

Requirements Traceability Matrix (RTM).
An initial version:

Requirement	User Story	Use Case	Test
FR-012	US-003	UC-008	TBD
FR-016	US-003	UC-008	TBD
FR-019	US-003	UC-009	TBD
FR-022	US-004	UC-010	TBD
FR-025	US-009	UC-011	TBD
FR-026	US-005	UC-013	TBD
FR-028	US-002	UC-007	TBD
FR-030	US-010	UC-014	TBD
FR-036	US-010	UC-016	TBD
FR-041	US-013	UC-015	TBD

Note: The TBD entries will be filled later in testing phase.

24. Requirements Validation

Correctness

Does each requirement represent a genuine need for transcript generation and academic record management?

Unambiguity

Can each requirement be interpreted in only one way without confusion?

Completeness

Does each requirement contain sufficient detail for implementation and testing?

Consistency

Are requirements free from conflicts with each other and with business rules?

Verifiability

Can we test whether each requirement has been satisfied through specific test cases?

Feasibility

Can each requirement be realistically implemented within the project constraints?

Traceability

Can we identify the origin of each requirement and follow it through to implementation?

Prioritization

Do we clearly understand which requirements are critical versus desirable?

Modifiability

Can requirements be updated without disrupting the entire specification?

25. Example of Requirement Quality
Poor:
The system should calculate grades quickly and accurately.

This is vague, not measurable, and lacks specific detail about what calculations are needed.

Better:
The system shall automatically calculate the Semester Grade Point Average (SGPA) for each completed semester using the formula: SGPA = Σ(Grade Points × Credits) / Σ(Credits), and display the result accurate to two decimal places.

This is specific, testable, and provides clear calculation logic.

26. Requirement Change Management

Because TGS follows Agile development, requirements can change during the project.

For example:

Change Request: Add email notification when a transcript is approved and ready for download.

The team should follow:

Change Request → Impact Analysis → Requirement Update → Backlog Update → Prioritization →
Sprint Planning → Design Change → Implementation → Testing → Release

The change should also update relevant documentation and the traceability matrix.

27. Future Requirements

Potential future requirements include:

1. bulk upload of grade records via file (CSV/Excel);
2. email/SMS notifications for status changes;
3. digital signature or QR-based verification on the transcript PDF;
4. integration with external universities or degree-verification agencies;
5. mobile application;
6. advanced analytics for institute management;
7. support for multiple academic programmes/curricula.

28. SRS Baseline and Versioning

- **Initial baseline:** SRS v1.0
- **Version Control Information:** This version establishes the core requirements for the Transcript Generation System, covering grade record management, automated SGPA and CGPA calculation, transcript generation with approval workflow, and basic reporting capabilities for academic office staff.

The LNM Institute of Information Technology, Jaipur

Software Development Laboratory

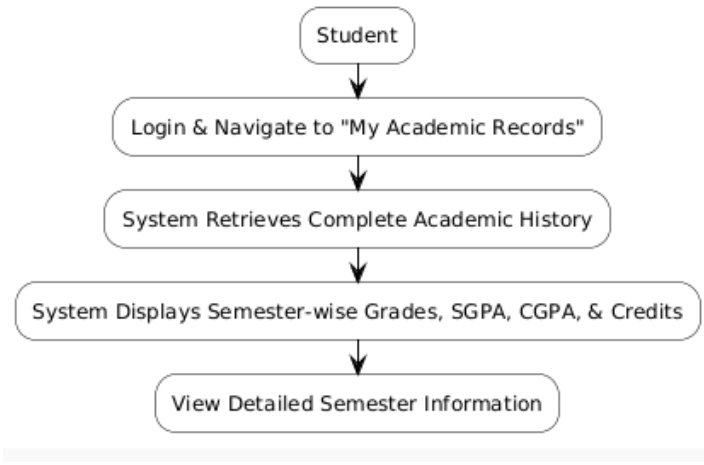
Course Code: CS314

Lab Assignment 3: Solution

30.Scenarios

The three detailed use cases carried are used below.

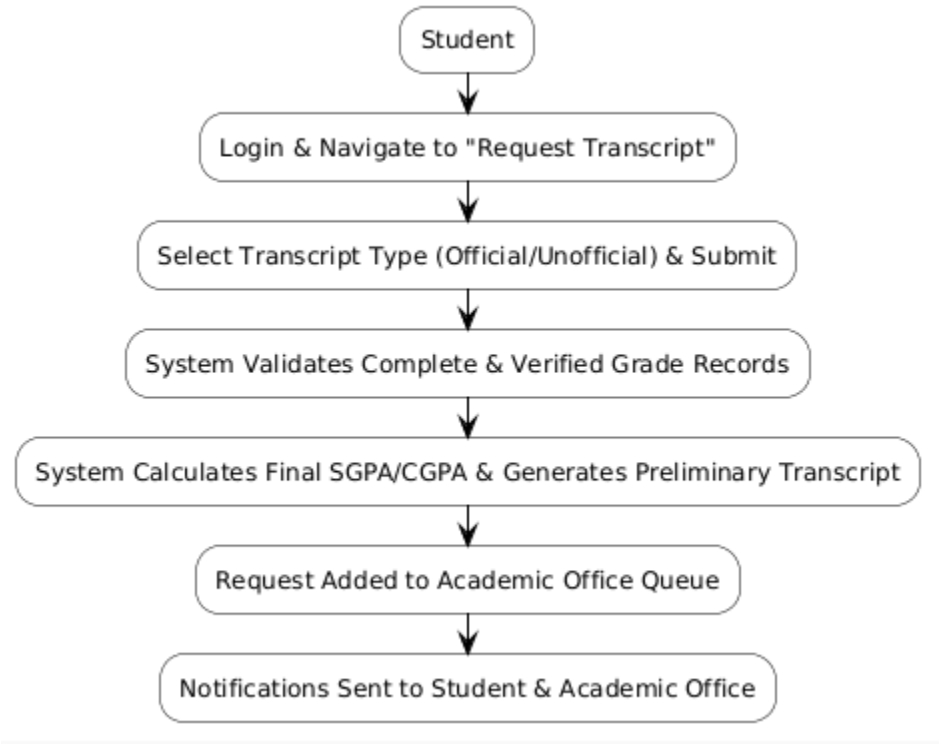
Scenario: Scenario: UC-007 - View Academic Records

Actor	Student
Goal	Student views their complete academic history, including semester-wise grades, calculated SGPA, and overall CGPA.
Pre-condition	Student is authenticated; student has at least one semester with grade records.
Main Flow	1. Student navigates to the "My Academic Records" section. 2. System retrieves the student's complete academic history. 3. System displays semester-wise grade records (course code, name, credits, and grade). 4. System displays calculated SGPA for each semester. 5. System displays overall CGPA and total credits earned. 6. Student views detailed information for any semester.
Representation	

Alternative / Exception Flow	• **No Records:** If no grade records exist, the system displays a message stating that no records are available. • **Pending Calculations:** If SGPA/CGPA calculation is pending, the system displays the status as "Under Processing".
Post Condition	Student has successfully viewed their complete academic records.

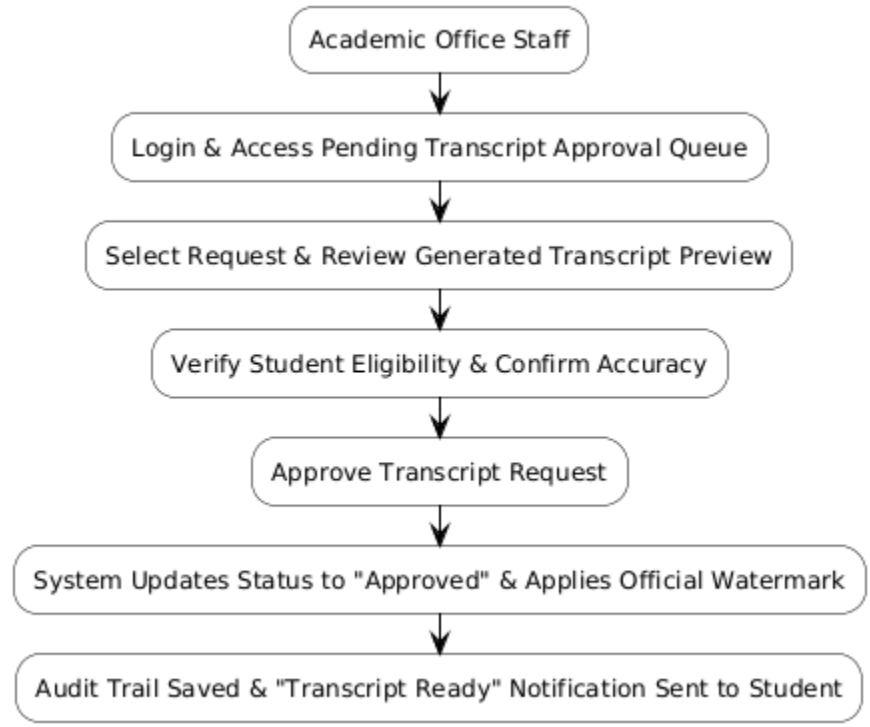
Scenario: UC-010 - Request Transcript

Actor	Student
Goal	Student requests an official or unofficial academic transcript for higher education or job opportunities.
Pre-condition	Student is authenticated; student has at least one completed semester with verified grades.
Main Flow	1. Student navigates to the "Request Transcript" section. 2. System displays the transcript request form. 3. Student selects transcript type (Official/Unofficial) and optionally provides a purpose. 4. Student submits the transcript request. 5. System validates that grade records are complete and verified. 6. System creates a transcript request record and calculates final SGPA and CGPA. 7. System generates a preliminary transcript and sends the request to the academic office for approval. 8. System displays confirmation to the student and notifies the academic office.

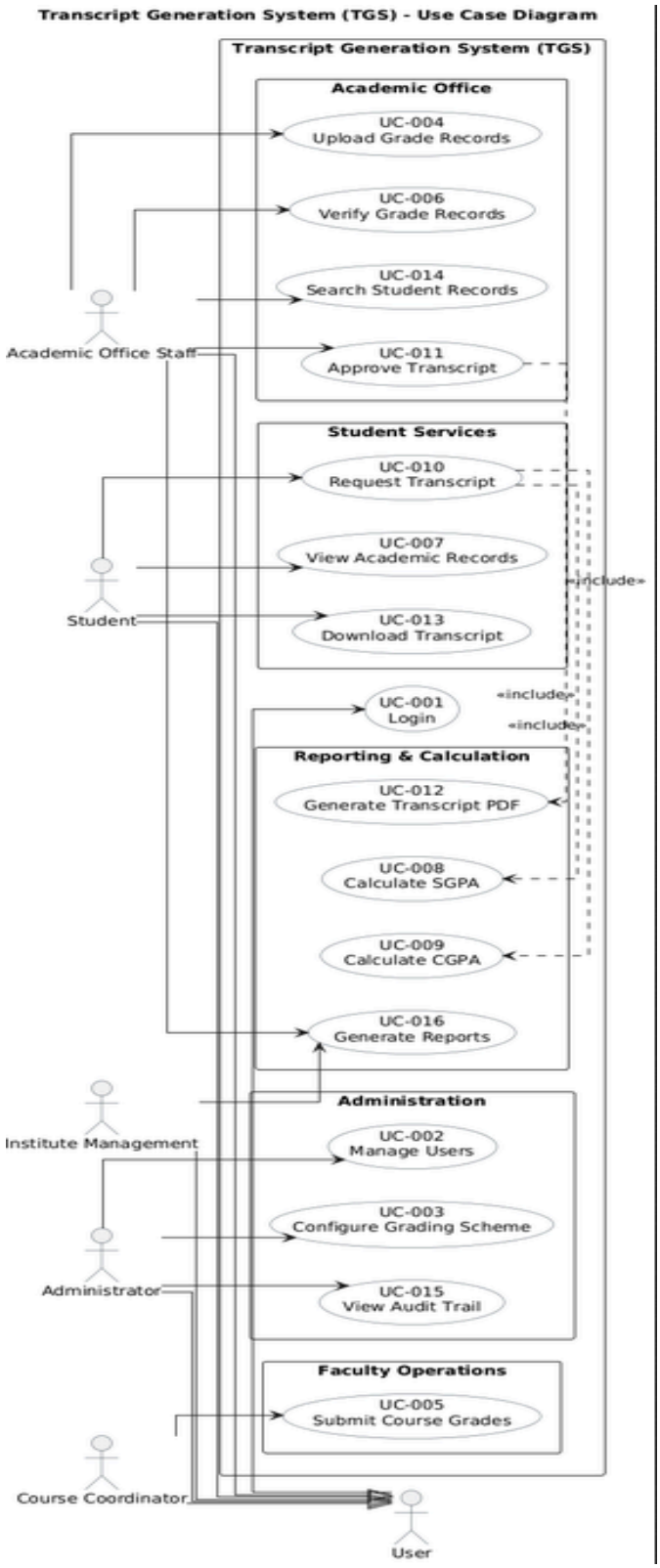
Representation	
Alternative / Exception Flow	• **Incomplete Records:** If grade records are incomplete or have pending corrections, the system blocks the request and notifies the student to resolve issues first.
Post Condition	A transcript request is successfully created, logged, and pending approval from the academic office.

Scenario: UC-011 - Approve Transcript

Actor	Academic Office Staff
Goal	Academic Office Staff reviews, verifies, and approves a pending transcript request, marking it official.
Pre-condition	Staff member is authenticated; a transcript request exists and is pending approval; all grade records are verified.

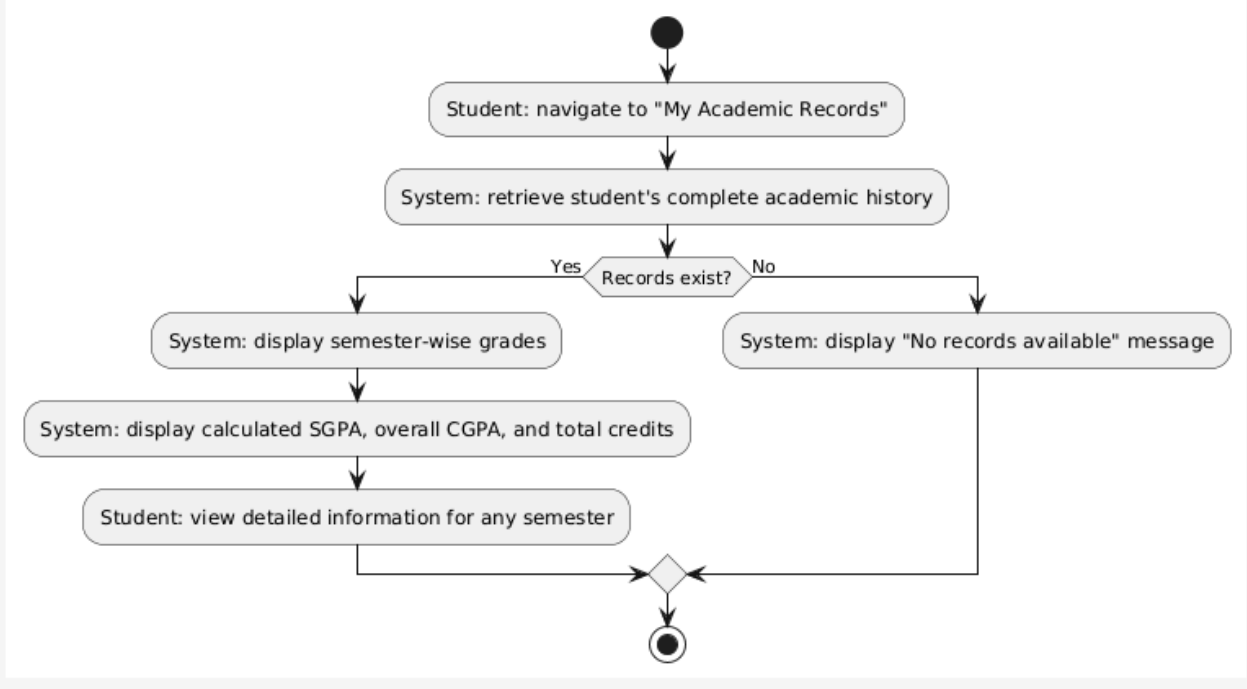
Main Flow	1. Staff member accesses the pending transcript approval queue. 2. System displays the list of pending transcript requests. 3. Staff member selects a transcript request to review. 4. System displays student details and a generated transcript preview. 5. Staff member reviews the transcript for accuracy and verifies student eligibility. 6. Staff member approves the transcript request. 7. System updates transcript status to "Approved", applies the official watermark, records approval details in the audit log, and notifies the student.
Representation	
Alternative / Exception Flow	• **Discrepancies Found:** If discrepancies are found or the student is not eligible, the staff member rejects the request with comments. The system updates the status and notifies the student.
Post Condition	Transcript is officially approved, marked with a watermark, logged in the audit trail, and ready for student download.

31. Use Case Diagram

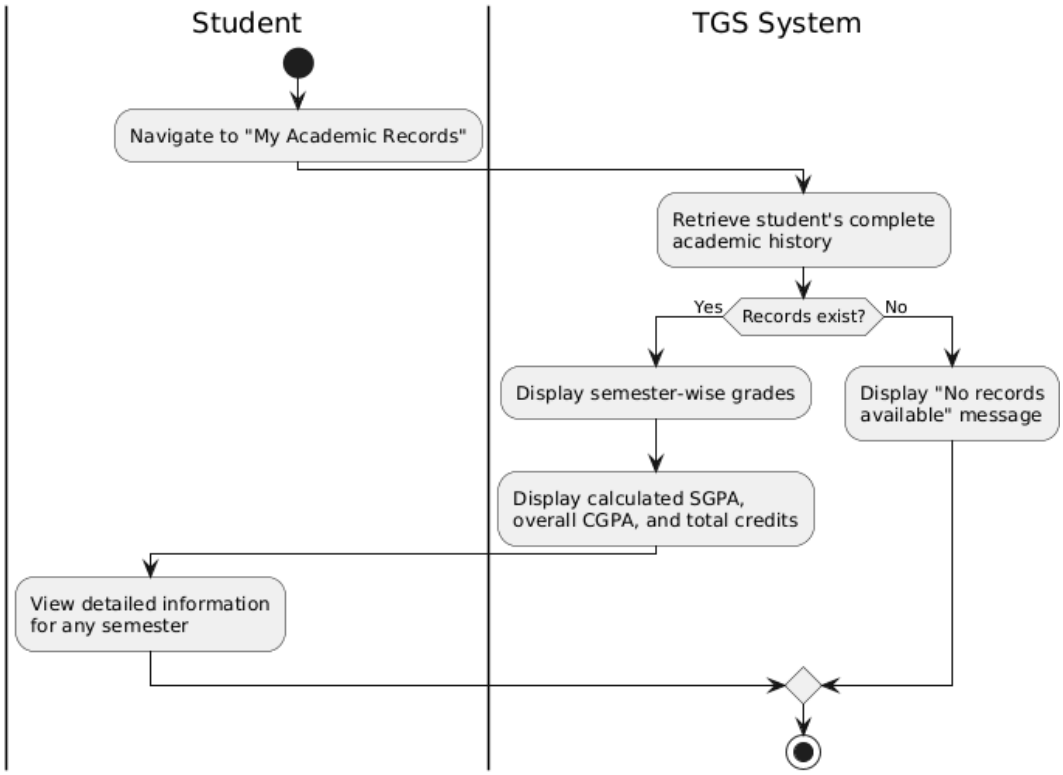


32. Activity Diagrams

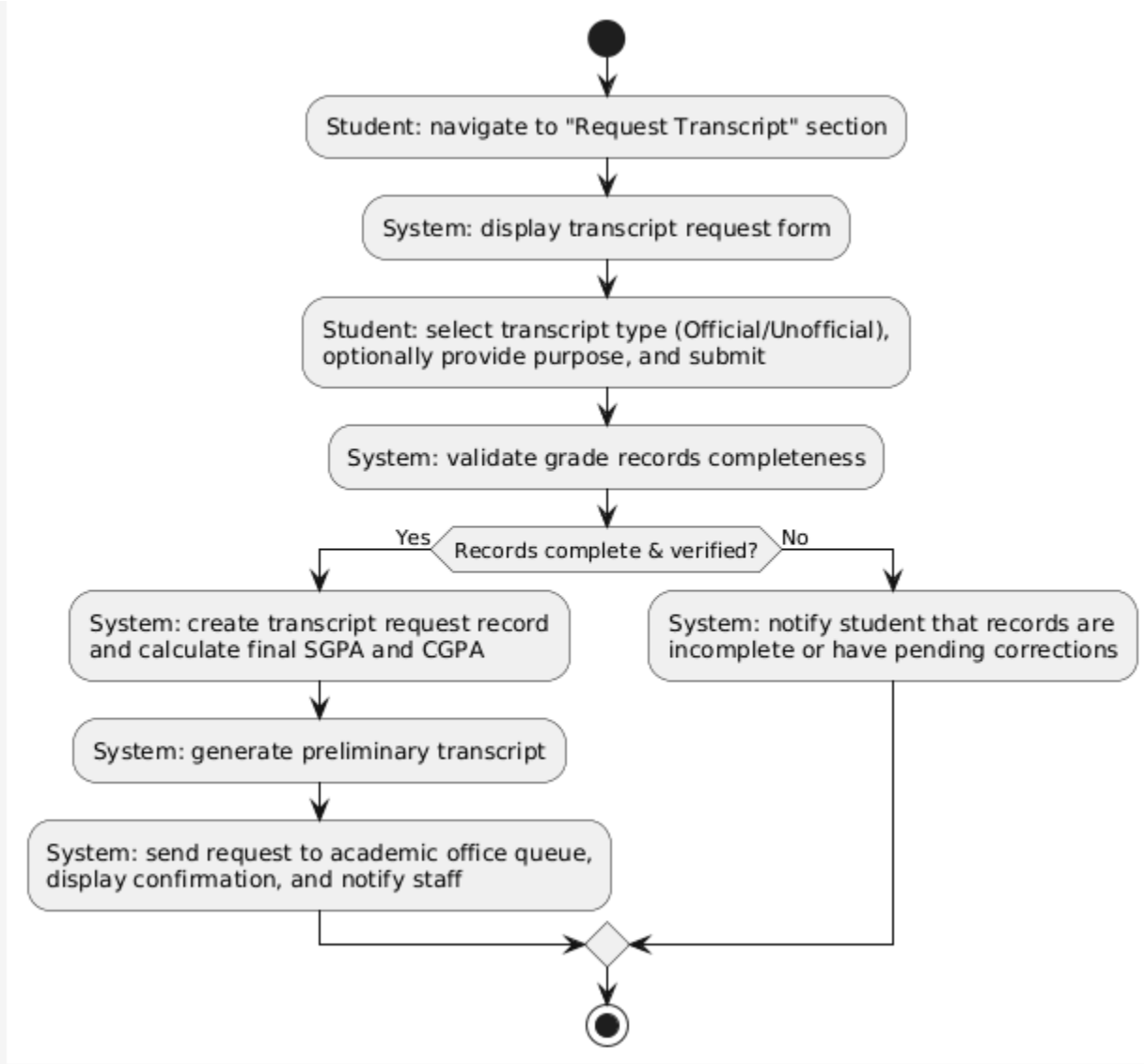
32.1 UC-007 - View Academic Records



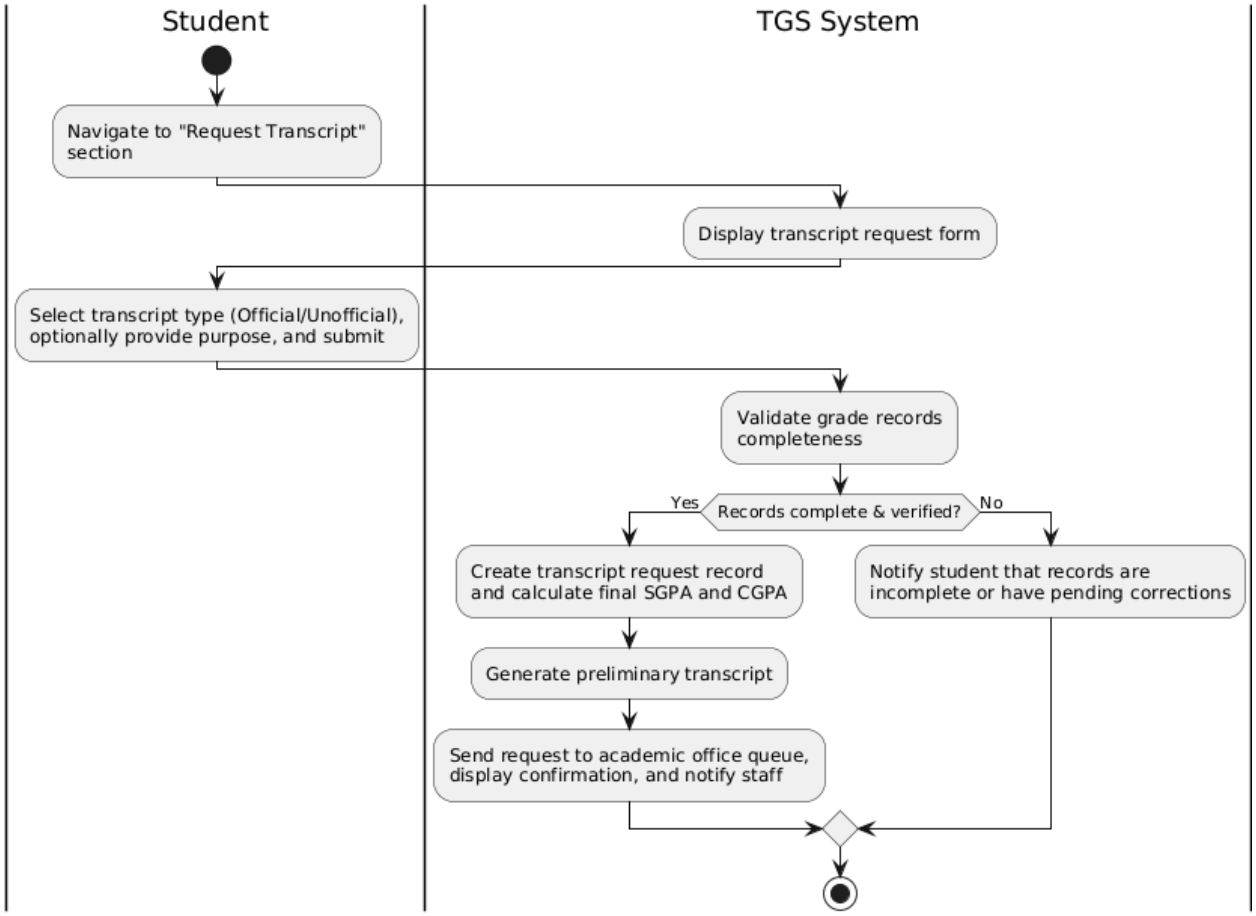
Swim Lanes



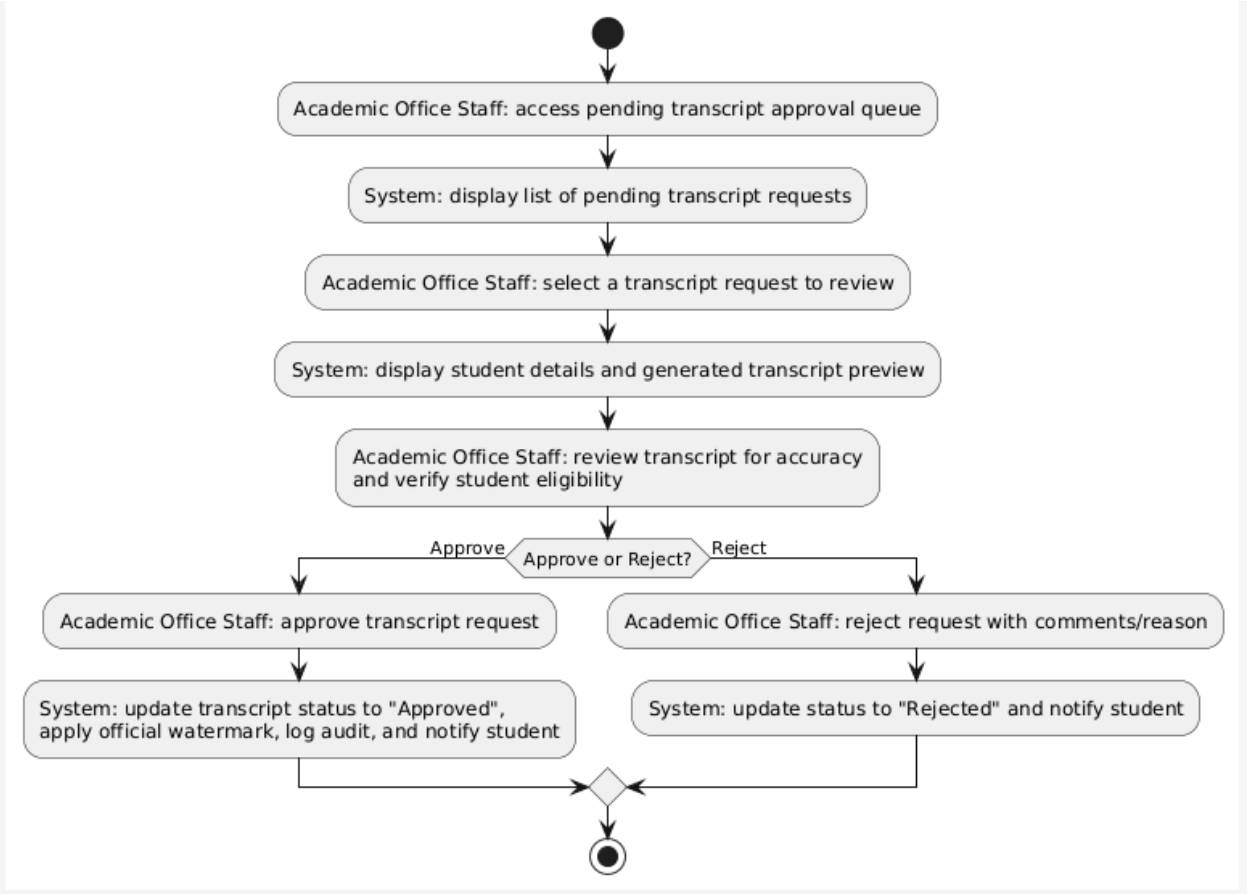
32.2 UC-010 - Request Transcript



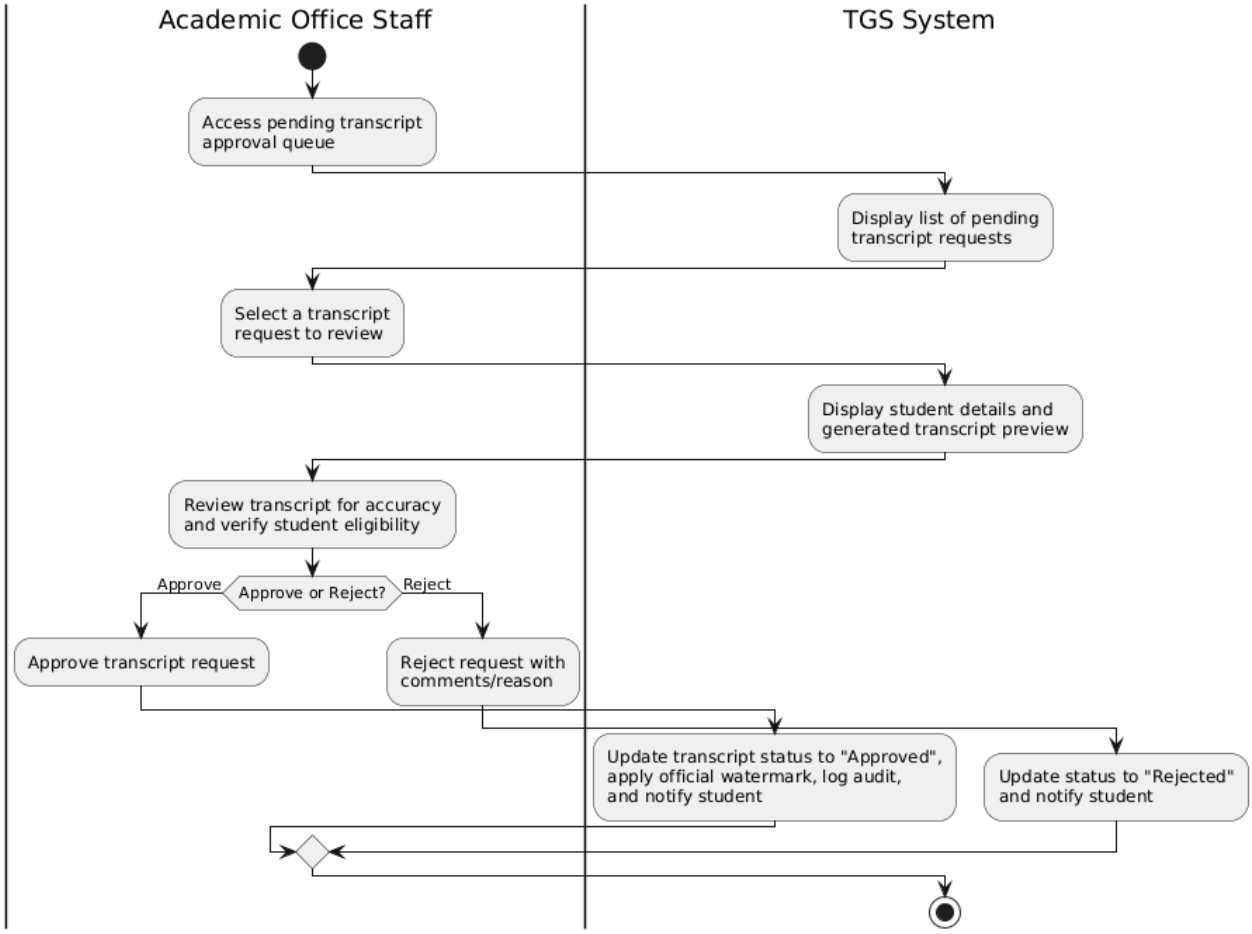
Swim Lanes


Student
TGS System
Navigate to "Request Transcript" section
Display transcript request form
Select transcript type (Official/Unofficial), optionally provide purpose, and submit
Validate grade records completeness
Yes
Records complete & verified?
No
Create transcript request record and calculate final SGPA and CGPA
Notify student that records are incomplete or have pending corrections
Generate preliminary transcript
Send request to academic office queue, display confirmation, and notify staff

32.3 UC-011 - Approve Transcript



Swim Lanes


Academic Office Staff
TGS System
Access pending transcript approval queue
Display list of pending transcript requests
Select a transcript request to review
Display student details and generated transcript preview
Review transcript for accuracy and verify student eligibility
Approve
Approve or Reject?
Reject
Approve transcript request
Reject request with comments/reason
Update transcript status to "Approved", apply official watermark, log audit, and notify student
Update status to "Rejected" and notify student

The LNM Institute of Information Technology, Jaipur

Software Development Laboratory

Course Code: CS314

Lab Assignment 4: Solution

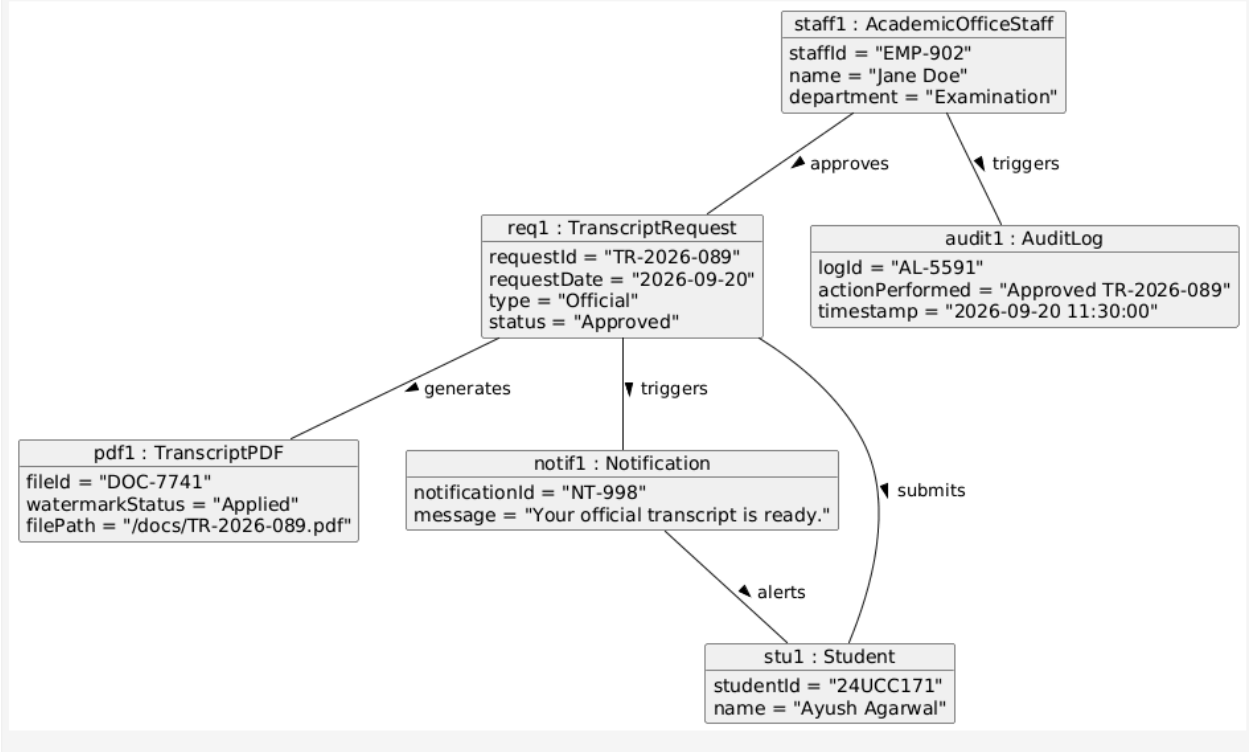
33. Classes

Eleven major classes were identified to support the Transcript Generation System (TGS). Note that the Notification system and Audit Log act as system components and are represented as classes rather than actors.

Class	Important Attributes	Important Operations / Responsibilities	Traceability
User	userID, name, email, role, accountStatus, lastLogin	authenticate(), logout()	FR-001–FR-003
Student	(inherits User), studentID, program, batch, currentSemester	viewAcademicRecords(), requestTranscript(), downloadTranscript()	FR-022, FR-026, FR-028, UC-007, UC-010
Academic OfficeStaff	(inherits User), staffID, department	sverifyGradeRecord(), approveTranscript(), searchStudentRecords()	FR-010, FR-025, FR-030, UC-011
Course	courseCode, courseName, credits, department, courseType	getCourseDetails()	FR-008
GradeRecord	recordID, studentID, courseCode, semester, gradeObtained, creditsEarned, verificationStatus	calculateGradePoints(), updateStatus()	FR-008, FR-012, BR-002, BR-003
SemesterRecord	recordID, studentID, semesterNumber, academicYear, totalCredits, sgpa, status	calculateSGPA(), addGradeRecord()	FR-016, FR-017, BR-004
AcademicHistory	historyID, studentID, cgpa, calculationDate	calculateCGPA(), getAllSemesterRecords()	FR-019, FR-020, BR-005
TranscriptRequest	transcriptID, studentID, generationDate, type, purpose, status, approvedBy	generatePreliminary(), approve(), reject()	FR-022, FR-025, BR-006, BR-012
TranscriptPDF	fileID, transcriptID, creationDate, watermarkStatus, filePath	applyWatermark(), generatePDF()	FR-024, FR-027
AuditLog	logID, userID, actionPerformed, entityAffected, changesMade, timestamp lastUpdatedBy	logAction()	FR-041, FR-042

Notification	notificationID, recipientID, notificationType, message, timestamp, readStatus	sendNotification()	FR-032–FR-034

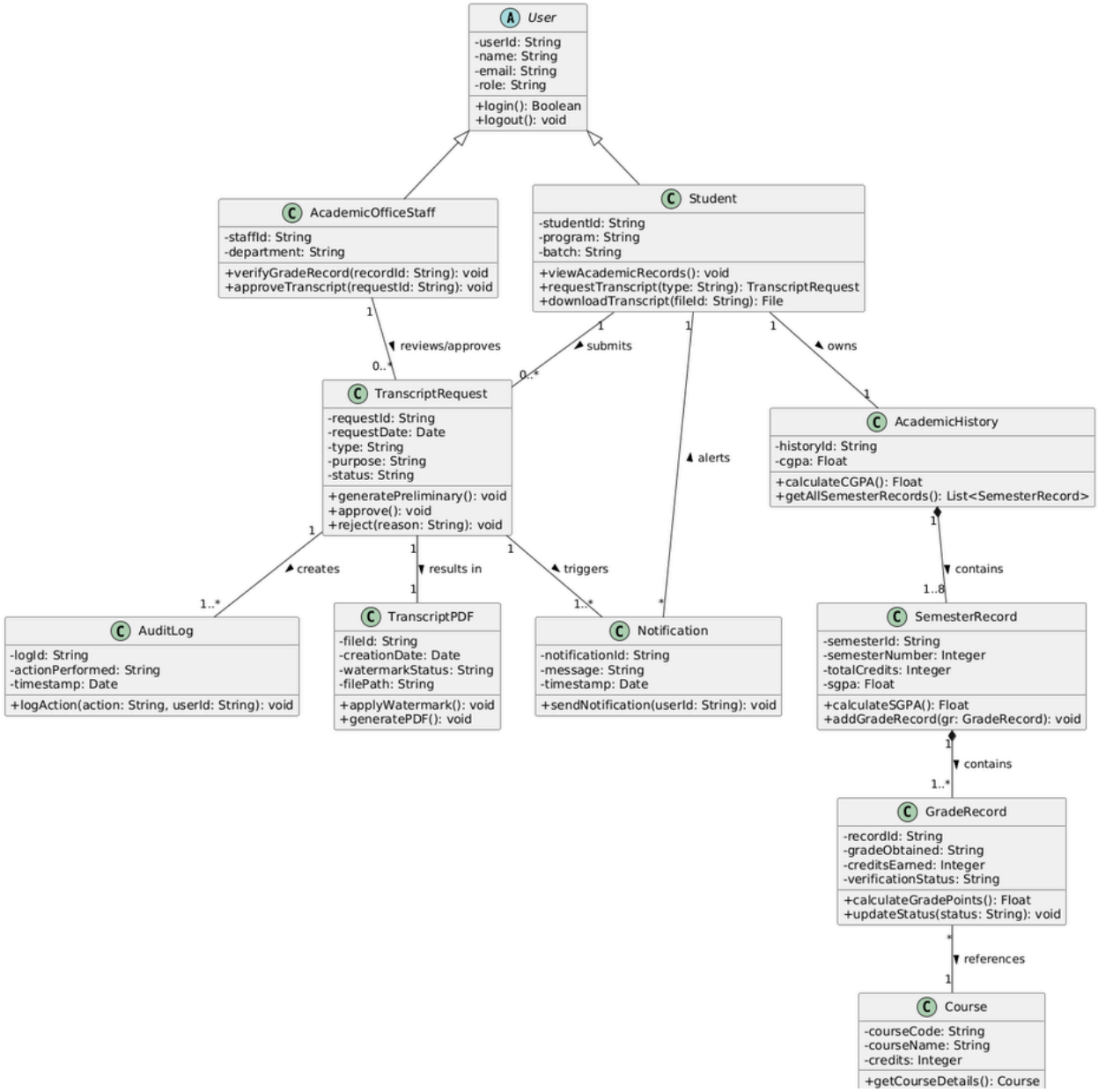
34. Object Diagram



Scenario Represented by the Object Diagram

This object diagram represents a meaningful snapshot of the system during the **UC-011: Approve Transcript** scenario. An instance of `AcademicOfficeStaff` (`staff1`) is actively reviewing and approving a pending `TranscriptRequest` (`req1`) submitted by a `Student` (`stu1`). Upon successful approval, the system links the request to the finalized `TranscriptPDF` (`pdf1`) with an official watermark applied. Simultaneously, the system triggers the creation of an `AuditLog` entry (`audit1`) for accountability and dispatches a `Notification` (`notif1`) alerting the student that their document is ready.

35. Class Diagram and Relationships



Justification of Relationships

Relationship	Type	Multiplicity	Justification
Student, AcademicOfficeStaff → User	Generalization	—	All roles authenticate identically (FR-001); only their permitted operations differ based on their role (FR-005).
AcademicHistory ◆— SemesterRecord	Composition	1 — 1..*	CGPA calculation includes all completed semesters (BR-005); a semester record does not exist independently of a student's history.

SemesterRecord ◆— GradeRecord	Composition	1 — 1..*	SGPA is calculated from specific grade records (FR-016, BR-004); a grade record has no independent lifecycle outside its semester.
Student — AcademicHistory	Association	1 — 1	A student is authorized to view their complete academic history including all semesters (FR-028).
GradeRecord — Course	Association	* — 1	Grade records are tied to specific courses to determine the course credit weight for calculations (FR-012).
Student — TranscriptRequest	Association	1 — 0..*	Authenticated students can request the generation of their academic transcript (FR-022, UC-010).
AcademicOfficeStaff — TranscriptRequest	Association	1 — 0..*	Academic office staff review and approve pending transcript generation requests (FR-025, UC-011).
TranscriptRequest ◆— TranscriptPDF	Composition	1 — 0..1	The system generates a standardized PDF for the request (FR-024); the document is meaningless if the request is deleted.
User — AuditLog	Association	1 — 0..*	Significant user actions, such as approvals and modifications, are logged against the acting user's identity (FR-041, FR-042).
User — Notification	Association	1 — 0..*	The system sends status and approval notifications to specific users (FR-033, FR-034).
TranscriptRequest ⇢ Notification	Dependency (<>)	—	Approving a transcript request directly triggers the creation of a notification sent to the student (UC-011).

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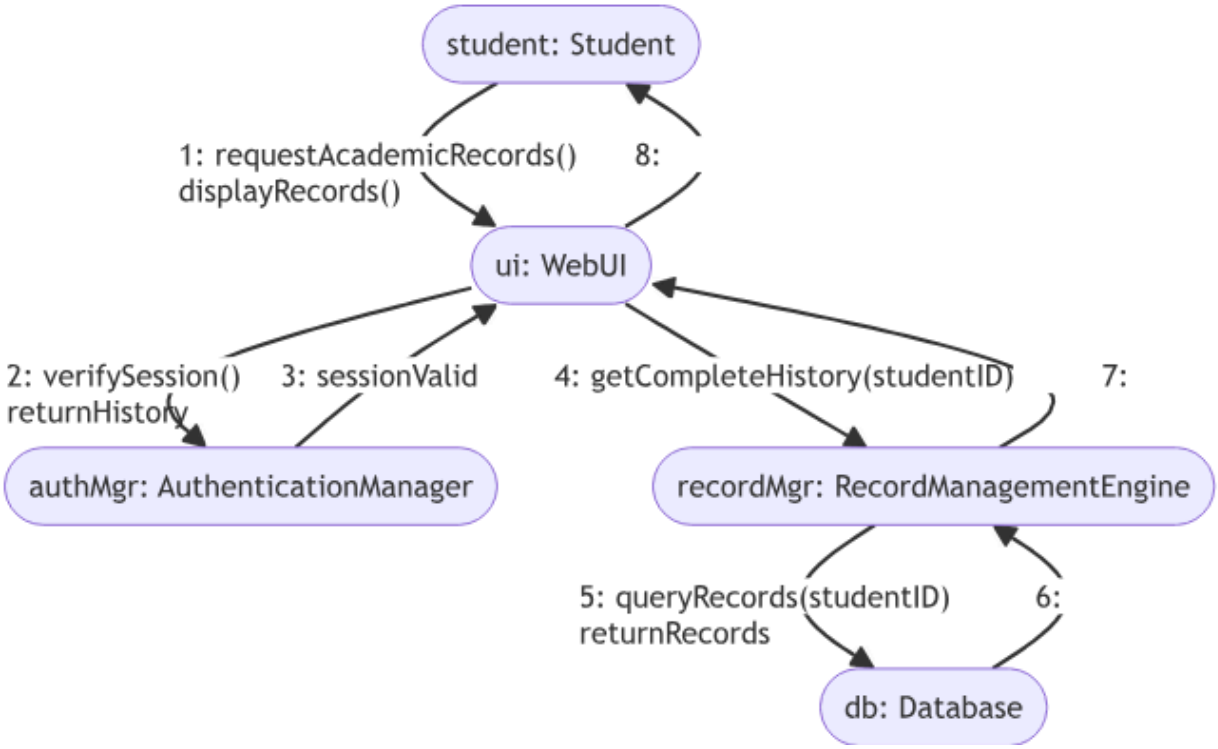
Course Code: CS314

Lab Assignment 5: Solution

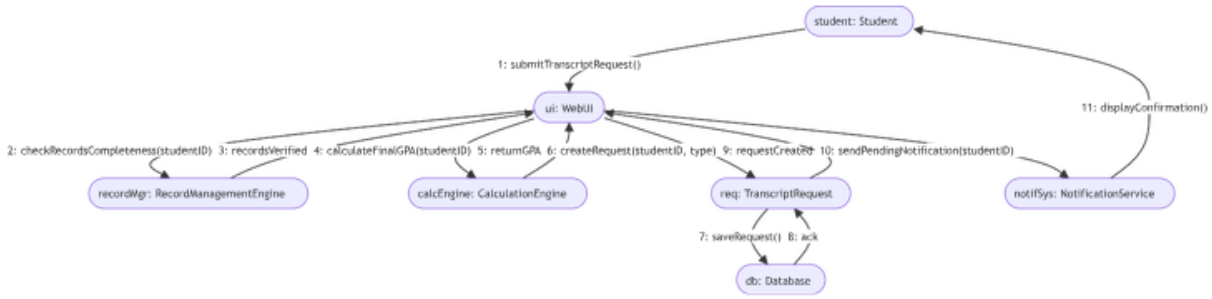
36. Collaboration Diagrams

The following collaboration diagrams model the object interactions for the three detailed use cases using flowcharts with message sequences to represent the collaboration relationships.

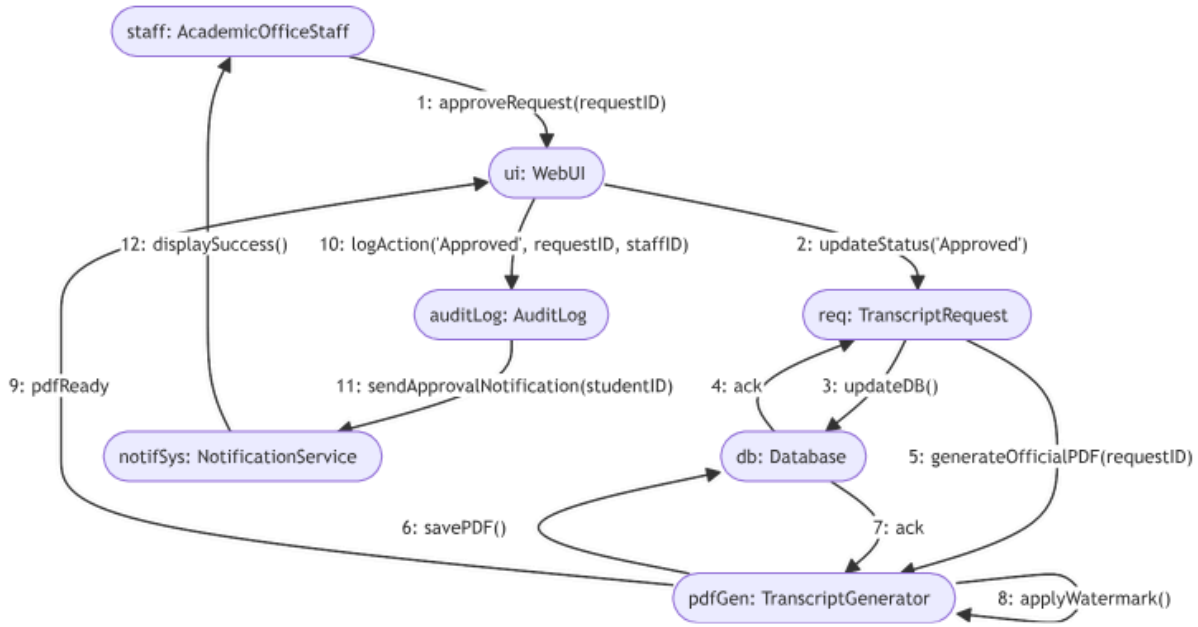
UC-007 - View Academic Records



UC-010 - Request Transcript

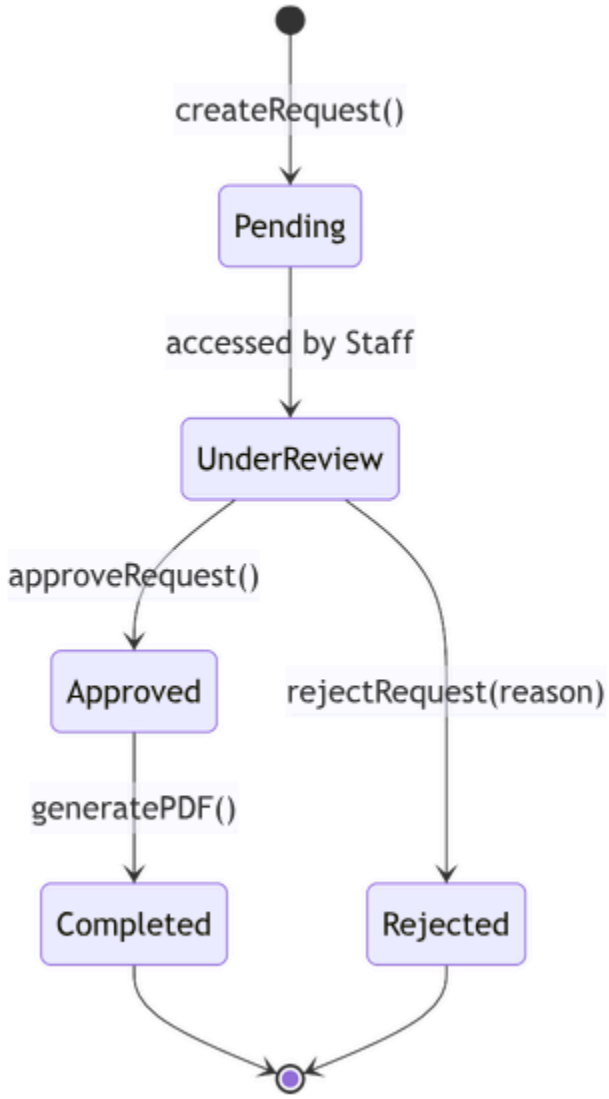


UC-011 - Approve Transcript

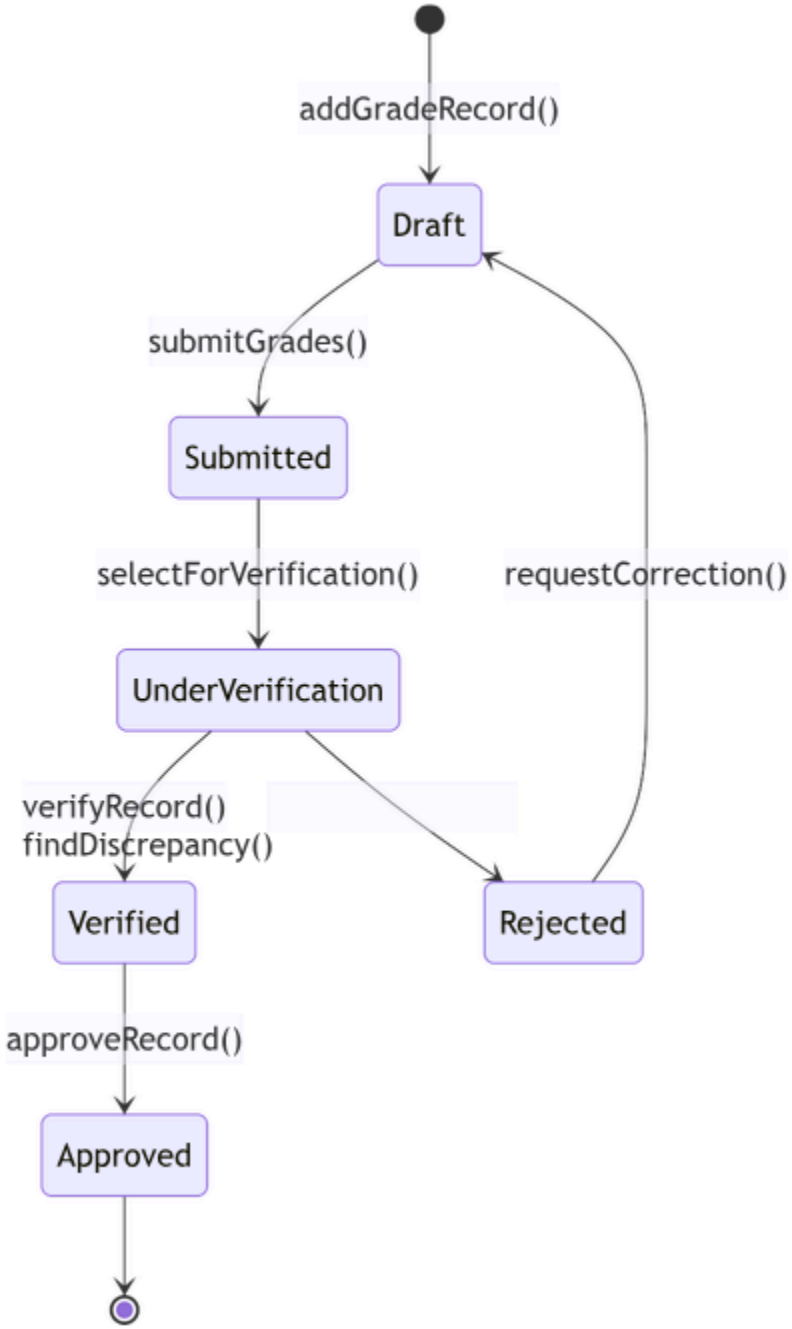


37. State Diagrams

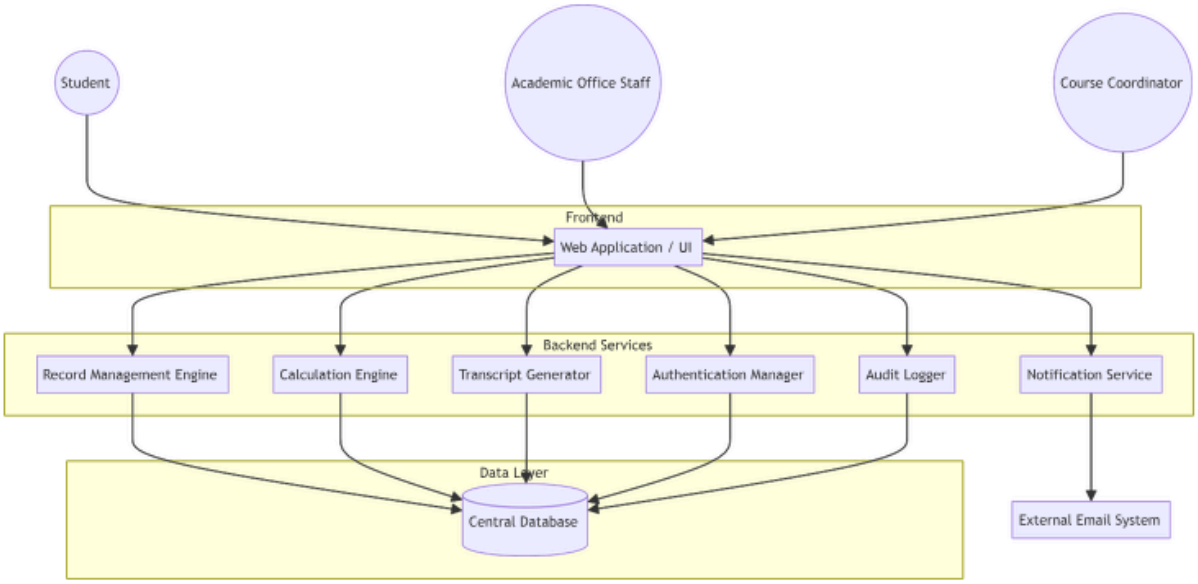
37.1 State Diagram for TranscriptRequest



37.2 State Diagram for SemesterRecord



38.Component Diagram



39. Individual Contribution

- **DAKSH KEDIA (24UCC174) – Project/Requirement Lead**

Ensured structural compliance by tracing all identified classes and associations directly back to the Functional Requirements and Business Rules established in the initial requirements specification. For Assignment 5, ensured end-to-end requirement traceability across the collaboration diagrams for UC-007, UC-010, and UC-011.

- **YASH SHARMA (24UCC172) – Design/Architecture Lead**

Designed and modeled the comprehensive UML Class Diagram, establishing the structural architecture, multiplicities, and specific generalization and composition relationships. For Assignment 5, designed the system Component Diagram across the Frontend, Backend Services, and Data Layer modules.

- **AYUSH AGARWAL (24UCC171) – Development Lead**

Identified the core system classes along with their essential attributes and operations, and designed the UML Object Diagram representing the transcript approval snapshot. For Assignment 5, modeled the UML State Diagrams for the complete lifecycles of TranscriptRequest and SemesterRecord.

- **JATIN DANGI (24UCC173) – Testing & Documentation Lead**

Documented the relationship justifications, verified architectural consistency between Assignment 3 activity diagrams and the new class structures, and managed the GitHub Markdown compilation. For Assignment 5, verified consistency between class models and collaboration interactions, and maintained report compilation alongside the AI policy declarations.

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AI USAGE POLICY – STUDENT DECLARATION

AI tools may be used as supporting tools for this assignment. The team is responsible for the correctness, consistency and understanding of all submitted requirements.

We declare that:

- We have disclosed the use of AI tools, if any.
- AI-generated suggestions have been reviewed, verified and adapted by the team.
- We understand and can explain the submitted requirements, use cases and user stories.
- We have verified that the requirements are consistent with our project's scope.
- The final SRS represents our team's understanding and decisions.

AI Tool(s) Used: Gemini (Google)

Purpose of AI Use: Supporting the team in structuring the requirements specification document, organizing functional and non-functional requirements based on our project scope, reviewing consistency between calculation logic and business rules, improving clarity and testability of requirements statements, and ensuring comprehensive coverage of transcript generation workflows. All AI suggestions have been reviewed, verified, and adapted by the team.

Team ID: T029

Team Members: AYUSH AGARWAL, YASH SHARMA, JATIN DANGI, DAKSH KEDIA

Date: 29 September 2026

Team Declaration:
We confirm that the information provided above is accurate and that we accept responsibility for the submitted work.